

A incerteza quando o mundo muda

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Sumário

Como este livro mede	5
1 A medida, o corte e o preço	7
1.1 Uma pergunta de todo dia	7
1.2 O dado e a regra	7
1.3 A promessa parece cumprida	8
1.4 O corte assina a entrega	8
1.5 O que a média não vê	9
1.6 O preço não desaparece: muda de endereço	10
1.7 O que fica	11

Como este livro mede

Este livro não afirma nada sem mostrar como se mede. Cada número que aparece no texto sai de um caderno de experimento: um caderno por medição, executável sozinho, com os parâmetros no topo para quem quiser mexer. O caderno termina gravando o resultado num arquivo, e cada figura sai duas vezes — em papel, para este livro, e em imagem, para inspeção.

O texto então não escreve o número: ele cita o comando que o laboratório gerou. Isso parece detalhe e não é: um número digitado à mão pode divergir da medição sem que ninguém perceba, e um número citado não pode, porque não existe em dois lugares.

A medição de abertura serve de exemplo. Ela lê uma série longa de preços e mede a volatilidade em duas escalas de tempo: 0,1925 ao longo de todo o histórico e 0,1299 no último ano, uma razão de 0,6749 em 6718 dias úteis. A figura mostra as duas curvas que produzem esses números.

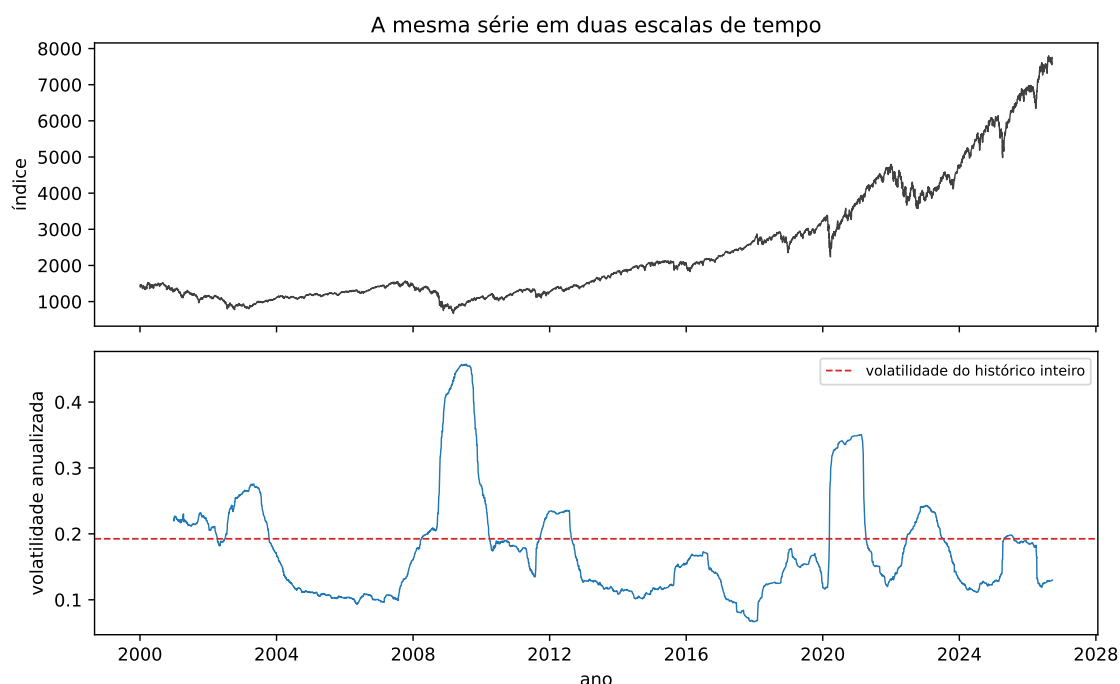


Figure 1: Preço e volatilidade da mesma série, em duas escalas de tempo. A linha tracejada é a volatilidade do histórico inteiro; a curva azul é a janela móvel de um ano.

Capítulo 1

A medida, o corte e o preço

1.1 Uma pergunta de todo dia

Quem carrega dinheiro aplicado faz, quase toda noite, a mesma pergunta: *quanto eu posso perder amanhã?* Não é pergunta de adivinho. É a pergunta que decide se a posição cabe no bolso de quem a carrega — e posição que não cabe no bolso é desmontada antes de o mercado explicar por quê.

Toda tentativa de responder tem a mesma forma, e vale a pena escrevê-la em partes, porque é dessas partes que este livro inteiro trata.

Primeiro escolhe-se **quantos dias olhar**. Depois olha-se o que aconteceu nesses dias e **põe-se uma linha** num dos piores. Uma regra dessas, escrita por inteiro, soa assim: *o corte de hoje é o décimo terceiro pior dia dos últimos 252 pregões*. E o que ela pretende valer também se anuncia: *não passo daqui, dezenove dias em vinte* — isto é, 5% de violação.

Quatro palavras, com o mesmo sentido daqui em diante:

- a **janela** é o pedaço do passado que a regra olha;
- o **corte** é onde a linha é posta dentro da janela;
- a **promessa** é a taxa de violação que se anuncia;
- a **entrega** é a taxa que de fato acontece.

A regra não é invenção deste livro: a linha posta no quantil dos piores dias é a medida de risco mais usada em finanças, discutida e corrigida há décadas [88]. A pergunta deste capítulo é estreita, e é a primeira de todas: *a entrega tem alguma coisa a ver com a promessa?*

1.2 O dado e a regra

O dado sobre o qual tudo aqui é medido é o retorno de um dia para o outro. Com o preço de hoje p_t e o de amanhã p_{t+1} ,

$$r_{t+1} = \log \frac{p_{t+1}}{p_t}.$$

A razão do logaritmo aparece no exemplo mínimo: um preço que desce e volta ao lugar tem retornos que somam zero.

$$100 \rightarrow 95 : -0,0513; \quad 95 \rightarrow 100 : +0,0513; \quad \text{queda de 5\% pede alta de 5,26\%}.$$

Em porcentagem a volta também acontece, mas com números diferentes, porque porcentagem multiplica onde o logaritmo soma. Somar, em vez de multiplicar, é o que permite acompanhar o dia a dia sem perder o fio.

A série é o índice S&P 500.

São 6718 pregões, de 4 de janeiro de 2000 a 21 de setembro de 2026. Ela vem do arquivo do projeto anterior, e o empréstimo fica declarado aqui — o dado atravessa, e todo número tirado dele é remedido neste capítulo.

Sobre ela, a regra tem duas decisões, e só duas. A **janela**: os 252 pregões anteriores a hoje. O **corte**: o décimo terceiro pior deles. O corte de cada dia é erguido com os dias que já passaram e com nenhum outro, de modo que a linha que se vê no gráfico estava lá antes de o dia acontecer. Um corte que espiasse o futuro acertaria sempre e não valeria nada.

1.3 A promessa parece cumprida

Aplicada aos 6718 pregões, a regra produz um corte novo a cada dia e uma resposta por dia: o dia ficou abaixo da linha, ou não ficou. Nos 6466 dias em que havia janela suficiente, a linha foi rompida 332 vezes.

$$\text{entrega} = 5,135\%, \quad \text{promessa} = 5\%.$$

A diferença é de um décimo de ponto percentual. Parece que funcionou.

Não funcionou, e o motivo não está no mercado: está na aritmética de erguer a linha.

1.4 O corte assina a entrega

Proposição 1.1 (a conta do corte). *Suponha que os n dias da janela e o dia seguinte sejam $n + 1$ valores de uma mesma lei contínua, em qualquer ordem igualmente provável. Se o corte é o k -ésimo menor dos n dias da janela, então a probabilidade de o dia seguinte ficar abaixo do corte é $\frac{k}{n + 1}$.*

Prova. Junte os $n + 1$ valores e ponha-os em ordem crescente. Todas as ordens são igualmente prováveis, então a posição do dia seguinte nessa fila é uniforme entre as $n + 1$ posições. O dia seguinte fica abaixo do corte exatamente quando ocupa uma das k primeiras posições, que são as ocupadas pelos k menores da janela. Logo a probabilidade é $k/(n + 1)$. $\square$

A lei ser contínua serve para que dias empatados tenham probabilidade zero, e a hipótese de que todas as ordens são igualmente prováveis é a hipótese de que os dias não têm nada a dizer uns sobre os outros — que é justamente o que um mundo que muda viola. A proposição não é a previsão do que vai acontecer; é o contraste contra o qual o que aconteceu se lê.

A proposição é um caso pequeno e exato de um viés conhecido: quantil estimado de uma amostra não entrega a probabilidade que se anuncia, e o tamanho do erro depende do tamanho da amostra [13].

Com $n = 252$ e $k = 13$, a conta dá $13/(252 + 1)$: 5,138%. Esse número não vem do mercado. Vem da decisão de olhar 252 dias e cortar no décimo terceiro. E a promessa anunciada está fora do alcance desse corte — o posto teria de cair entre dois dias, e não existe o décimo terceiro dia e meio de um ano. Com 252 pregões, as duas taxas ao lado da promessa são 4,743%, no décimo segundo, e 5,138%, no décimo terceiro. Nenhuma das duas é 5%.

Medimos, e o mercado entregou 5,135% — 332 rompimentos em 6466 dias. É o que o corte assinou, com poucos milésimos de diferença. Para saber se essa diferença diz alguma

coisa, falta o único contraste que responde: submeter um mundo em que nada muda exatamente à mesma rotina.

Sorteamos 200 séries do tamanho da nossa, com retornos independentes e sempre com a mesma lei, e aplicamos a cada uma a mesma regra. A entrega média desses mundos foi $5,139 \pm 0,1091$ — contra os 5,138% da conta. E o mundo real entregou 5,135%.

A média não distingue o mundo real de um mundo que nunca mudou.

Este é o primeiro princípio do livro, e ele é desconfortável: a média de violações é uma **propriedade do corte, não do mundo**. Não porque o mundo não apareça, mas porque o corte já assinou o número antes de qualquer dado chegar. Quem lê só a média lê a própria escolha e acredita estar lendo o mercado.

1.5 O que a média não vê

O mundo aparece em outro lugar, e o lugar é a *forma* com que as violações chegam: espalhadas pelo calendário, ou concentradas em pedaços. Para medir a forma, contamos quantas violações caem em cada janela móvel de 60 dias — a isso chamamos um **bloco** —, e olhamos o bloco mais cheio. A promessa de 5% admite 3 violações em 60 dias.

A mediana dos blocos é 2: na maior parte do tempo a linha aguenta mais do que prometeu. O pior bloco tem 20 violações, e termina em 20 de novembro de 2008. Onde a promessa admitia 3, aconteceram 20.

Nos 200 mundos que nunca mudam, o pior bloco de cada um teve, em média, 9,26 violações, e o pior bloco de todos eles chegou a 13. Nenhum passou disso. A série real, com o mesmo tamanho e a mesma rotina, chegou a 20.

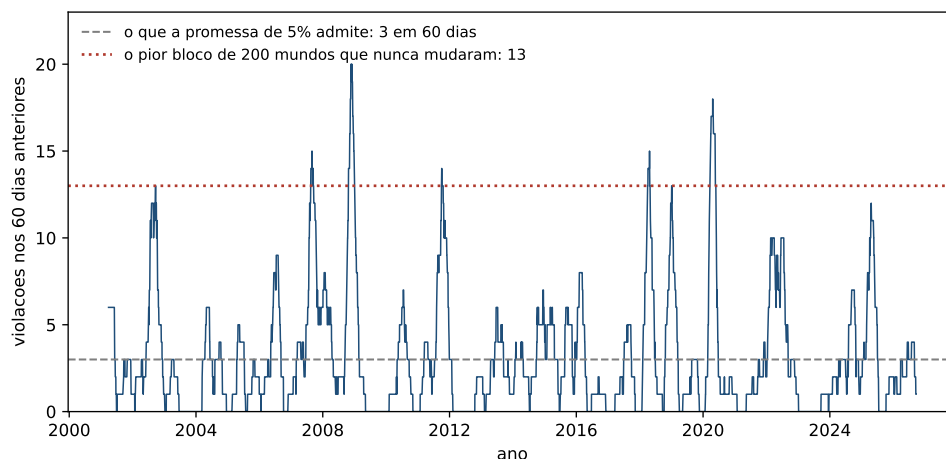


Figure 1.1: Violações nos 60 dias anteriores, dia a dia. A linha tracejada é o que a promessa de 5% admite; a pontilhada é o pior bloco que apareceu em 200 mundos que nunca mudam.

A figura mostra o que a média esconde: a série passa trechos longos colada na linha da promessa e sobe, em surtos curtos, até cruzar a linha pontilhada. E os blocos têm data. O teto dos mundos que nunca mudam é cruzado em 6 episódios distintos, que ocupam 1,951% dos blocos medidos, em 2007, 2008, 2011, 2018 e 2020. Nem todos esses anos têm nome de crise, e todos aconteceram neste mercado.

A leitura do bloco corrige ainda a primeira impressão: 13,05% dos blocos passam do dobro da promessa. O que se promete não é o que se entrega, nem na maior parte do tempo, nem nos piores pedaços — é uma média entre os dois.

1.6 O preço não desaparece: muda de endereço

Se o corte assina a média, uma saída parece evidente: escolher outra janela. Vale a pena tentar, e a tentativa é barata, porque é a mesma regra com outro número. Varremos a janela de um mês a cinco anos, mantendo a promessa anunciada de 5% e o bloco de 60 dias.

janela (pregões)	a conta do corte (%)	a entrega (%)	o pior bloco (violações)
21	9,091	9,407	12
63	6,25	6,612	15
252	5,138	5,135	20
1260	4,996	5,093	24

Table 1.1: A mesma regra com quatro janelas, a mesma promessa anunciada de 5% e o mesmo bloco de 60 dias. A coluna do meio é a conta da Teorema 1.1; a última é o pior bloco medido.

As três colunas contam uma história só, e ela é a resposta deste capítulo.

A **conta do corte** obedece à promessa anunciada apenas na janela longa. Na janela de um mês, o corte assina 9,091% — quase o dobro dos 5% anunciados —, e a razão é aritmética, não mercado: com poucos dias não existe corte que produza 5%.

A **entrega** acompanha a conta do corte nas quatro janelas, um pouco acima dela em todas. Esse resto é a contribuição do mundo, e é pequeno. Encurtar a janela não deixa a medida mais fina; torna a promessa mais caro do que se anunciou.

O **pior bloco** caminha no sentido oposto: 12, 15, 20, 24. Quanto mais longo o passado que a linha olha, mais tempo ela leva para acompanhar o que mudou, e mais violações cabem no pedaço ruim. Na janela de cinco anos o pior bloco chega a 24. E, no extremo oposto, uma janela curta não compra bloco nenhum: o pior bloco do mundo real, 12, fica *abaixo* do pior bloco de um mundo que nunca muda, porque uma janela curta se move junto com o mercado — e cobra o preço na média.

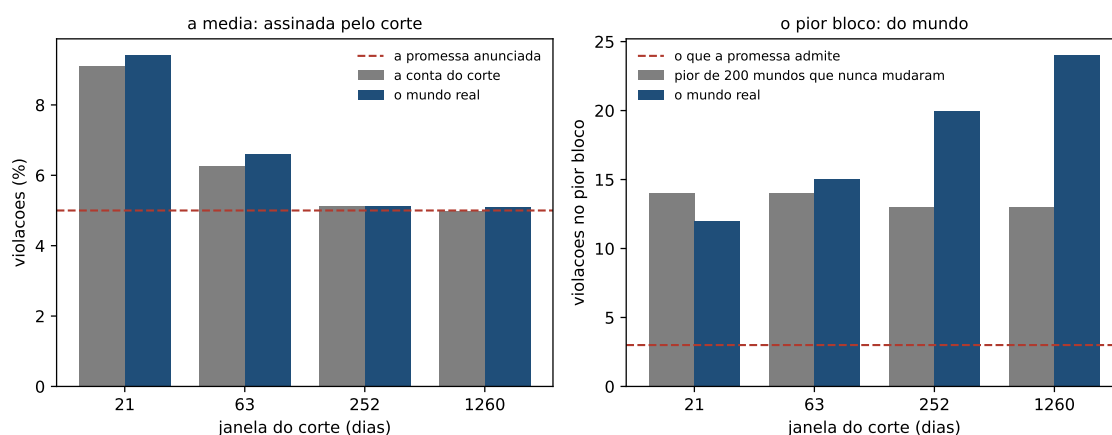


Figure 1.2: As duas leituras, nas quatro janelas. À esquerda, a média: a conta do corte e o mundo real andam juntos. À direita, o pior bloco: o mundo real contra o pior dos 200 mundos que nunca mudam.

O preço de fixar não desaparece: muda de endereço. Encurtar o corte cobra na média; alongá-lo cobra no bloco. Não há janela em que a linha entregue o que a promessa anunciou — e a razão é a Teorema 1.1, que não deixa o mundo entrar na conta.

1.7 O que fica

Ficam três movimentos, e eles são o instrumento deste livro. Primeiro, **declarar o corte**: a taxa que se assina não é a taxa que se anuncia, e a diferença é a conta da Teorema 1.1. Segundo, **ler em bloco, não em média**: a média é assinada pelo corte, e só o bloco responde pelo mundo. Terceiro, **varrer o corte**: o preço não some quando se troca de janela — ele se muda, e saber para onde ele foi é a decisão.

E fica um fracasso, visível e sem resposta aqui. Um corte fixo supõe que o pedaço de passado que ele olha se parece com o pedaço que vem. A série deste capítulo não se parece consigo mesma: passa meses colada na linha e depois rompe a linha 20 vezes em 60 dias. Se os blocos têm data, alguma coisa no mundo não é fixa, e a média não é o instrumento que a vê.

A próxima pergunta, então, não é qual janela escolher — *é que formas a mudança tem*: se ela chega pela média, pela cauda, pela dependência entre coisas que antes andavam separadas, ou pelo calendário. Responder exige um instrumento novo: um orçamento de erro declarado, contra o qual cada mudança possa ser cobrada. É para lá que este livro vai agora.

Fontes

- [1] G. A. Carpenter, S. Grossberg. A massively parallel architecture for a self-organizing neural pattern recognition machine. Boston University; Northeastern University, 1987.
- [2] Stephen Grossberg. Competitive Learning: From Interactive Activation to Adaptive Resonance. Boston University (CNS Tech Lab), 1987.
- [3] D. E. Rumelhart, D. Zipser. Feature discovery through competitive learning. University of California at San Diego, 1985.
- [4] T. J. Sejnowski, C. R. Rosenberg. NETtalk: A parallel-network that learns to read aloud. Johns Hopkins University, 1986.
- [5] Thomas Bayes, Richard Price. An Essay towards solving a Problem in the Doctrine of Chances. Royal Society, 1763.
- [6] Norman Dalkey, Olaf Helmer. An Experimental Application of the Delphi Method to the Use of Experts. RAND Corporation, 1962.
- [7] Judea Pearl. Bayesian Networks*. University of California, Los Angeles (UCLA), 2000.
- [8] Stefan Conrady, Lionel Jouffe. Bayesian Networks & BayesiaLab: A Practical Introduction for Researchers. BayesiaLab (Bayesia), 2015.
- [9] David Aebischer, John Vatterott, Jr., Michael Grimes, Andrew Vatterott, Roderick Jordan, Carlo Reinoso, Bradford Alex Baker, William D. Aldrich, Luis Reinoso, Rodolfo Villalba, Michael Johnson, Christopher Myers, Stefan Conrady, Joseph A. Tatman, Suzanne M. Mahoney, Darrin L. Whaley, Amanda B. Hepler. Bayesian Networks for Combat Equipment Diagnostics. U.S. Army Communications Electronics Command, Aberdeen, Maryland, 2017.
- [10] Stefan Conrady. Exchange-Traded Funds: Knowledge Discovery, Interpretation, and Anomaly Detection. Bayesia USA, 2018.
- [11] Andrea Fontanari, Pasquale Cirillo, Nassim Nicholas Taleb. Gini Estimation Under Infinite Variance. STEM Academic Press, 2020.
- [12] Nassim Nicholas Taleb, Michail Loulakis, Spyros Makridakis, Pasquale Cirillo. How Much Data Do You Need? An Operational Metric for Fat-Tailedness. STEM Academic Press, NYU Tandon School of Engineering, 2020.
- [13] Nassim Nicholas Taleb, Raphael Douady. On the Super-Additivity and Estimation Biases of Quantile Contributions. STEM Academic Press, 2020.

- [14] Nassim Nicholas Taleb, Brandon Yarckin, Chitpuneet Mann, Damir Delic, Mark Spitznagel. Option Pricing Under Power Laws: A Robust Heuristic. Universa Investments, STEM Academic Press, 2020.
- [15] Vladimir Vovk, Bin Wang, Ruodu Wang. Admissible ways of merging p-values under arbitrary dependence. University of London, University of Waterloo, 2022.
- [16] Hongjian Wang, Aaditya Ramdas. Anytime-valid t-tests and confidence sequences for Gaussian means with unknown variance. Carnegie Mellon University, 2025.
- [17] Vladimir Vovk, Ruodu Wang. E-values: Calibration, combination, and applications. University of London, University of Waterloo, 2021.
- [18] Ian Waudby-Smith, Aaditya Ramdas. Estimating means of bounded random variables by betting. Carnegie Mellon University, 2024.
- [19] Ruodu Wang, Aaditya Ramdas. False discovery rate control with e-values. University of Waterloo, Carnegie Mellon University, 2022.
- [20] Martin Larsson, Aaditya Ramdas, Johannes Ruf. The numeraire e-variable and reverse information projection. Carnegie Mellon University, London School of Economics, 2025.
- [21] Larry Wasserman, Aaditya Ramdas, Sivaraman Balakrishnan. Universal inference. Carnegie Mellon University, 2020.
- [22] Ilya Prigogine. La Fin des Certitudes. Editions Odile Jacob, 1996.
- [23] Ilya Prigogine; Isabelle Stengers. The End of Certainty. University of Texas at Austin; Free University of Brussels, 1997.
- [24] Matthias Seeger. Bayesian Gaussian Process Models: PAC-Bayesian Generalisation Error Bounds and Sparse Approximations. University of Edinburgh, 2003.
- [25] Matthias Seeger, Christopher Williams, Neil Lawrence. Fast Forward Selection to Speed Up Sparse Gaussian Process Regression. University of Edinburgh, 2003.
- [26] Neil Lawrence, Matthias Seeger, Ralf Herbrich. Fast sparse Gaussian process methods: The informative vector machine. Microsoft Research, Cambridge; University of Edinburgh, 2003.
- [27] Matthias Seeger. PAC-Bayesian Generalization Error Bounds for Gaussian Process Classification. University of Edinburgh, 2002.
- [28] Nassim Nicholas Taleb. Machine Learning Considerations. NYU Tandon School of Engineering, 2020.
- [29] Nassim Nicholas Taleb, Pasquale Cirillo, Raphael Douady, Andrea Fontanari, Hélyette Geman, Donald Geman, Espen Haug. Statistical Consequences of Fat Tails: Real World Preasymptotics, Epistemology, and Applications. NYU Tandon School of Engineering, Universa Investments, 2020.
- [30] Henri Bergson. Essai sur les données immédiates de la conscience Time and Free Will. Lycée de Clermont-Ferrand, 1888.
- [31] Henri Bergson. Matter and Memory Matière et mémoire. Collège de France, 1896.

- [32] Mark B. N. Hansen. *New Philosophy for New Media*. Princeton University, 2004.
- [33] Brian Massumi. *Parables for the Virtual*. Université de Montréal, 2002.
- [34] Gustav Fechner. *The Elements of Psychophysics*. Universität Leipzig (Leipzig University), 1860.
- [35] Shibhansh Dohare, J. Fernando Hernandez-Garcia, Qingfeng Lan, Parash Rahman, A. Rupam Mahmood, Richard S. Sutton. *Loss of Plasticity in Deep Continual Learning*. University of Alberta, Amii, 2024.
- [36] Shibhansh Dohare, Qingfeng Lan, A. Rupam Mahmood. *Overcoming Policy Collapse in Deep Reinforcement Learning*. University of Alberta, 2023.
- [37] V. Farias, A. D. Jozefiak. *Self-Normalized Resets for Plasticity in Continual Learning*. Massachusetts Institute of Technology (MIT), 2025.
- [38] G. Sokar, R. Agarwal, P. S. Castro, U. Evci. *The Dormant Neuron Phenomenon in Deep Reinforcement Learning*. Eindhoven University of Technology; Google DeepMind; Mila, 2023.
- [39] Yang Song, Stefano Ermon. *Generative Modeling by Estimating Gradients of the Data Distribution*. Stanford University, 2019.
- [40] Jaehyeok Shin, Aaditya Ramdas, Alessandro Rinaldo. *E-detectors: a nonparametric framework for sequential change detection*. Carnegie Mellon University, 2022.
- [41] Jonas Peters, Peter Bühlmann, Nicolai Meinshausen. *Causal inference using invariant prediction: identification and confidence intervals*. ETH Zürich, University of Copenhagen, 2016.
- [42] Bernhard Schölkopf, Francesco Locatello, Stefan Bauer, Nan Rosemary Ke, Nal Kalchbrenner, Anirudh Goyal, Yoshua Bengio. *Toward Causal Representation Learning*. Max Planck Institute for Intelligent Systems, Mila, Google Research, 2021.
- [43] Vincent Plassier, Mehdi Makni, Aleksandr Rubashevskii, Eric Moulines, Maxim Panov. *Conformal Prediction for Federated Uncertainty Quantification Under Label Shift*. Ecole Polytechnique (France), Lagrange Mathematics and Computing Research Center, MBZUAI (EAU), Technology Innovation Institute, 2023.
- [44] C. Dong, C. Smadi, V. A. Vatutin. *Critical branching processes in random environment and Cauchy domain of attraction*. Xidian University (China), Univ. Grenoble Alpes (France), Steklov Mathematical Institute (Russia), 2020.
- [45] Topi Halme, H. Vincent Poor, Visa Koivunen. *Adaptive Sequential Change Detection using Mixtures of Predictive Distributions*. Aalto University, Princeton University, 2026.
- [46] Tero Karras, Miika Aittala, Timo Aila, Samuli Laine. *Elucidating the Design Space of Diffusion-Based Generative Models (EDM)*. NVIDIA, 2022.
- [47] Kevin Black, Cheng Chi, Abhishek Gupta, Sergey Levine. *Training Diffusion Models with Reinforcement Learning (DDPO)*. UC Berkeley, 2023.

- [48] E. Lemaire; L. Cordone; A. Castagnetti; P.-E. Novac et al.. An analytical estimation of spiking neural networks energy efficiency. Citado em estudo da Delft University of Technology, 2023.
- [49] Enrique Barba Roque; Luis Cruz. Energy Aware Development of Neuromorphic Implantables: From Metrics to Action. Delft University of Technology (TU Delft), 2025.
- [50] J. Yik; K. V. den Berghe; D. den Blanken; Y. Bouhadjar et al.. Neurobench: A framework for benchmarking neuromorphic computing algorithms and systems. Harvard University; Delft University of Technology; Forschungszentrum Jülich; IMEC Netherlands; City University of Hong Kong; Intel; Istituto Italiano di Tecnologia (colaboração multi-institucional), 2025.
- [51] Aaditya Ramdas. Hypothesis testing using e-values, martingales & betting. Carnegie Mellon University, 2021.
- [52] Roberto Alvarez-Martinez e Pedro Miramontes. Early Warning Signals in Ecological Time-Series. Universidad Autónoma de Querétaro; UNAM; Universidad Autónoma de San Luis Potosí, 2026.
- [53] Mark Braverman e Or Zamir. Optimality of Frequency Moment Estimation. Princeton University; Tel Aviv University, 2025.
- [54] Pasquale Cirillo e Nassim Nicholas Taleb. Tail Risk of Contagious Diseases. Delft University of Technology; New York University (Tandon School of Engineering), 2020.
- [55] Shashank Gupta, Chaitanya Ahuja, Tsung-Yu Lin, Sreya Dutta Roy, Harrie Oosterhuis, Maarten de Rijke, Satya Narayan Shukla. A Simple and Effective Reinforcement Learning Method for Text-to-Image Diffusion Fine-tuning. University of Amsterdam, Meta AI, Radboud University, 2026.
- [56] Bernhard Schölkopf, Francesco Locatello, Stefan Bauer, Nan Rosemary Ke, Nal Kalchbrenner, Anirudh Goyal, Yoshua Bengio. Towards Causal Representation Learning. Max Planck Institute, Mila, Google Research, 2021.
- [57] James P Jun, Vijay Marupudi, Raj Sanjay Shah, Sashank Varma. A Neural Network Model of Complementary Learning Systems: Pattern Separation and Completion for Continual Learning. Georgia Institute of Technology, 2025.
- [58] Claudia Czado, Tilmann Gneiting, Leonhard Held. Predictive Model Assessment for Count Data. Technische Universität München, University of Washington, Universität Zürich, 2007.
- [59] David Woodruff. Optimal Space Lower Bounds for all Frequency Moments. MIT, CMU School of Computer Science, 2004.
- [60] Adam Paszke, Micha Pilipczuk. VC Density of Set Systems Definable in Tree-Like Graphs. University of Warsaw, 2020.
- [61] Andrei Nikolaevich Kolmogorov. Grundbegriffe der Wahrscheinlichkeitsrechnung. Moscow State University, 1933.
- [62] Preetha Ramiah, James Q. Smith, Silvia Liverani, Jamie Addison, Annabel Whipp, Oliver Bunnin. Bayesian Graphs of Intelligent Causation. Queen Mary University of London, University of Warwick, The Alan Turing Institute, Dstl, 2024.

- [63] Xi Chen, Yining Wang, Yu-Xiang Wang. Non-stationary Stochastic Optimization under $L_{\{p,q\}}$ -Variation Measures. Stern School of Business (NYU), Carnegie Mellon University, Amazon Inc., 2018.
- [64] Vladimir Vovk, Alex Gammerman, Glenn Shafer. Algorithmic learning in a random world. Royal Holloway (University of London), Rutgers School of Business, 2022.
- [65] Borja Sánchez-López, Jesus Cerquides. Dual Stochastic Natural Gradient Descent and convergence of interior half-space gradient approximations. IIIA-CSIC, Campus UAB, Bellaterra (Spain), 2021.
- [66] Juraj Bodik, Valérie Chavez-Demoulin. Structural restrictions in local causal discovery: identifying direct causes of a target variable. HEC, Université de Lausanne, 2025.
- [67] Suhas Thejaswi, Clemens Westrup. Approximate Computation of Frequency Moments. Aalto University, 2016.
- [68] Stephen Grossberg. A Path Toward Explainable AI and Autonomous Adaptive Intelligence: Deep Learning, Adaptive Resonance, and Models of Perception, Emotion, and Action. Center for Adaptive Systems, Boston University, 2020.
- [69] Velez e Clune. Diffusion-based neuromodulation can eliminate catastrophic forgetting in simple neural networks. Computer Science Department, University of Wyoming; Uber AI Labs, 2017.
- [70] Kirkpatrick et al.. Overcoming catastrophic forgetting in neural networks. Google DeepMind (United Kingdom); Imperial College London, 2017.
- [71] Song et al.. Consistency Models. OpenAI, 2023.
- [72] Kim et al.. Consistency Trajectory Models: Learning Probability Flow ODE Trajectory of Diffusion (CTM). KAIST, 2024.
- [73] Yin et al.. Improved Distribution Matching Distillation for Fast Image Synthesis (DMD2). MIT, Adobe, 2024.
- [74] Song & Dhariwal. Improved Techniques for Training Consistency Models (iCT). OpenAI, 2023.
- [75] Kaiwen Zheng, Yuji Wang, Qianli Ma, Huayu Chen, Jintao Zhang, Yogesh Balaji, Jianfei Chen, Ming-Yu Liu, Jun Zhu, Qinsheng Zhang. Large Scale Diffusion Distillation via Score-Regularized Continuous-Time Consistency (rCM). Alibaba (Wan 2.1), NVIDIA (Cosmos), 2025.
- [76] Yin et al.. One-step Diffusion with Distribution Matching Distillation (DMD). MIT, Adobe, 2024.
- [77] Lu & Song. Simplifying, Stabilizing and Scaling Continuous-Time Consistency Models (sCM / TrigFlow). OpenAI, 2024.
- [78] Jonathan Ansari, Ludger Rüschendorf. Sklar’s theorem, copula products, and ordering results in factor models. University of Freiburg, 2021.

- [79] Alexander Ly, Udo Boehm, Peter Grünwald, Aaditya Ramdas, Don van Ravenzwaaij. A Tutorial on Safe Anytime-Valid Inference: Practical Maximally Flexible Sampling Designs for Experiments Based on e-Values. Centrum Wiskunde & Informatica, Tilburg University, Leiden University, Carnegie Mellon, University of Groningen, 2023.
- [80] Albert-László Barabási, Réka Albert, Carlson, Doyle, Perdomo, Hardt, Peters, Bühlmann, Meinshausen. *Redes Complexas, Criticidade e Transições de Regime: Uma Síntese Rigorosa de Topologia, Dinâmica de Bifurcação, Performatividade e Invariância Causal*. s.d..
- [81] Marten Scheffer, Stephen R. Carpenter. *Catastrophic regime shifts in ecosystems: linking theory to observation*. Wageningen University, University of Wisconsin, 2003.
- [82] Ralf Herbrich, Thore Graepel. *A PAC-Bayesian Margin Bound for Linear Classifiers: Why SVMs work*. Technical University of Berlin, 2000.
- [83] PabloAMC, Jaime Sevilla, Victor Veitch, Claudia Shi, Zhijing Jin. *Causal representation learning as a technique to prevent goal misgeneralization*. AI Alignment Forum, University of Oxford, DeepMind, Columbia University, Max Planck, ETH, UC Berkeley, 2023.
- [84] Simon Bing, Jonas Wahl, Urmi Ninad, Jakob Runge. *Invariance & Causal Representation Learning: Prospects and Limitations*. Technische Universität Berlin, German Aerospace Center (DLR), 2023.
- [85] Zhaoyu Zhu, Rui Gao, Shuang Li. *Wasserstein Policy Gradient for Entropy-Regularized Linear-Quadratic Control*. University of Illinois Urbana-Champaign, University of Texas at Austin, Chinese University of Hong Kong, 2026.
- [86] Julien Bastian, Benjamin Leblanc, Pascal Germain, Amaury Habrard, Christine Largeron, Guillaume Metzler, Emilie Morvant, Paul Viallard. *PAC-Bayesian Generalization Guarantees for Fairness on Stochastic and Deterministic Classifiers*. Université Jean Monnet Saint-Étienne, Université Laval, Univ Rennes, 2026.
- [87] Annalisa Di Clemente, Claudio Romano. *Calibrating and Simulating Copula Functions in Financial Applications*. Sapienza University of Rome, 2021.
- [88] Ryan O’Connell. *Extreme Value Theory in Finance: Tail Risk, GEV, and GPD*. Ryan O’Connell Finance LLC, 2026.
- [89] Haoze Li, Jun Xie. *Feature Matching Intervention: Leveraging Observational Data for Causal Representation Learning*. Purdue University, 2025.
- [90] Juan C. Perdomo, Tijana Zrnic, Celestine Mendler-Dünner, Moritz Hardt. *Performative Prediction*. University of California, Berkeley, 2020.
- [91] Xu, Xie. *Wasserstein-Regularized Conformal Prediction under General Distribution Shift*. The Hong Kong University of Science and Technology (Guangzhou); Harbin Institute of Technology; Lehigh University, 2025.
- [92] Ah-hwee Tan. *Adaptive resonance associative map*. National University of Singapore / Singapore Management University, 1995.

- [93] Takaaki Koike, Marius Hofert, Haruki Tsunekawa. Measuring multivariate maximal tail dependence. Hitotsubashi University, University of Hong Kong, 2026.
- [94] Samantha Fairchild, Max Goering, Christian Wei. FAMILIES OF WELL APPROXIMABLE MEASURES. University of Washington, Ruhr West University of Applied Sciences, 2021.
- [95] Yi-Yu Lin. Sklar’s Theorem and Quantum State Reconstruction from All-Context Dependence. Fudan University, Shanghai Institute for Mathematics and Interdisciplinary Sciences (SIMIS), 2026.
- [96] Zhanglu Yan, Zhenyu Bai, Kaiwen Tang, Weng-Fai Wong. Reconsidering the Energy Efficiency of Spiking Neural Networks Inference from Analytical Perspectives. National University of Singapore (NUS), 2026.
- [97] Carlos Sebastián, Carlos E. González-Guillén, Jesús Juan. Enhancing reliability in prediction intervals using point forecasters: Heteroscedastic Quantile Regression and Width-Adaptive Conformal Inference. Fortia Energía, Universidad Politécnica de Madrid, 2025.
- [98] Alison L. Gibbs, Francis Edward Su. On choosing and bounding probability metrics. York University, Harvey Mudd College, 2002.
- [99] Omar Rivasplata, Ilja Kuzborskij, Csaba Szepesvári, John Shawe-Taylor. PAC-Bayes Analysis Beyond the Usual Bounds. University College London, DeepMind, Amii, 2020.
- [100] Kevin Black, Michael Janner, Yilun Du, Ilya Kostrikov, Sergey Levine. Training Diffusion Models with Reinforcement Learning. University of California, Berkeley; MIT, 2024.
- [101] J. Choi, J. Choi, M. Kang. Generative modeling through the semi-dual formulation of unbalanced optimal transport. Seoul National University, 2023.
- [102] P. Mokrov, A. Korotin, L. Li, A. Genevay, J. M. Solomon, E. Burnaev. Large-scale wasserstein gradient flows. Skolkovo Institute of Science and Technology, MIT, 2021.
- [103] Jaemoo Choi, Jaewoong Choi, Myungjoo Kang. Scalable Wasserstein Gradient Flow for Generative Modeling through Unbalanced Optimal Transport. Seoul National University, Korea Institute for Advanced Study, 2024.
- [104] Ulrike von Luxburg, Bernhard Schölkopf. Statistical Learning Theory: Models, Concepts, and Results. Max Planck Institute for Biological Cybernetics, 2008.
- [105] Glenn Shafer, Vladimir Vovk. The Sources of Kolmogorov’s Grundbegriffe. Rutgers Business School, Royal Holloway University of London, 2006.
- [106] E. de Amo, M. Díaz Carrillo, J. Fernández-Sánchez. Characterization of all copulas associated with non-continuous random variables. University of Almería, University of Granada, 2012.
- [107] Vladimir Vapnik, Alexey Chervonenkis. Vapnik–Chervonenkis dimension. 2026.
- [108] Simon Pope, Audun Jsang. Analysis of Competing Hypotheses using Subjective Logic. The University of Queensland, DSTC Pty Ltd, 2005.

- [109] Ryan J. Tibshirani, Rina Foygel Barber, Emmanuel J. Candès, Aaditya Ramdas. Conformal Prediction Under Covariate Shift. Stanford University, University of Chicago, Carnegie Mellon University, 2019.
- [110] Rainer Dahlhaus. The Estimation of Parameter Curves for Locally Stationary Processes. Universität Heidelberg, 1997.
- [111] Audun Jsang. Cumulative and Averaging Fission of Beliefs. UniK, University of Oslo, 2007.
- [112] Matthias Seeger. Bayesian Gaussian Process Models: PAC-Bayesian Generalization Error Bounds and Sparse Approximations. University of Edinburgh, 2003.
- [113] Motonobu Kanagawa, Philipp Hennig, Dino Sejdinovic, Bharath K. Sriperumbudur. Gaussian Processes and Kernel Methods: A Review on Connections and Equivalences. University of Tübingen; Max Planck Institute for Intelligent Systems; University of Oxford; Pennsylvania State University, 2018.
- [114] Jeremy Bernstein, Alex Farhang, Yisong Yue. Kernel Interpolation as a Bayes Point Machine. Caltech, 2022.
- [115] Matthias Seeger. PAC-Bayesian generalization bounds for gaussian processes. University of Edinburgh (Division of Informatics), 2002.
- [116] Matthieu Garcin, Maxime L. D. Nicolas. Nonparametric estimator of the tail dependence coefficient: balancing bias and variance. Léonard de Vinci Pôle Universitaire, Université Paris I Panthéon-Sorbonne, 2026.
- [117] Kwonho Kim, Heejeong Nam, Inwoo Hwang, Sanghack Lee. Towards Causal Representation Learning with Observable Sources as Auxiliaries. Seoul National University, Brown University, Columbia University, 2025.
- [118] Alexander Mey, Rui Manuel Castro. Invariant Causal Prediction with Locally Linear Models. Eindhoven University of Technology (TU/e), ASML Research, 2024.
- [119] Willem Anna Mark Wybo. Using the Green’s function to simplify and understand dendrites. EPFL (École Polytechnique Fédérale de Lausanne), 2017.
- [120] Wilfredo Palma, Ricardo Olea. An efficient estimator for locally stationary Gaussian long-memory processes. Pontificia Universidad Católica de Chile, 2010.
- [121] Rainer Dahlhaus. Locally stationary processes. Universität Heidelberg, 2012.
- [122] Rainer Dahlhaus, Wolfgang Polonik. Nonparametric quasi maximum likelihood estimation for Gaussian locally stationary processes. Universität Heidelberg, University of California, 2006.
- [123] Rainer Dahlhaus, L. Giraitis. On the optimal segment length for parameter estimates for locally stationary time series. Universität Heidelberg, 1998.
- [124] Wilfredo Palma, Ricardo Olea, Guillermo Ferreira. Estimation and forecasting of locally stationary processes. Sociedad Chilena de Estadística, 2013.
- [125] Linbo Wang, Thomas Richardson, James Robins. Causal Inference: A Tale of Three Frameworks. University of Toronto, University of Washington, Harvard T.H. Chan School of Public Health, 2025.

- [126] Rabanus Derr, Robert C. Williamson. Systems of Precision: Coherent Probabilities on Pre-Dynkin-Systems and Coherent Previsions on Linear Subspaces. University of Tübingen, Tübingen AI Centre, 2023.
- [127] Frank Lad, Giuseppe Sanfilippo, Gianna Agrò. Extropy: Complementary Dual of Entropy. University of Canterbury (NZ), University of Palermo (Italy), 2015.
- [128] AmirEmad Ghassami, Saber Salehkaleybar, Negar Kiyavash, Kun Zhang. Learning Causal Structures Using Regression Invariance. University of Illinois at Urbana-Champaign, Carnegie Mellon University, 2017.
- [129] Macaluso Girolamo, Mandelli Lorenzo, Mirko Bicchierai, Stefano Berretti, Andrew D. Bagdanov. No MoCap Needed: Post-Training Motion Diffusion Models with Reinforcement Learning using Only Textual Prompts. University of Florence, 2025.
- [130] Lihua Lei, Emmanuel J. Candès. Conformal Inference of Counterfactuals and Individual Treatment Effects. Stanford University, 2021.
- [131] Badr-Eddine Chérif-Abdellatif, Pierre Alquier, Mohammad Emtiyaz Khan. A Generalization Bound for Online Variational Inference. CREST, ENSAE, Institut Polytechnique de Paris; RIKEN Center for AI Project, 2019.
- [132] Allen Z. Ren, Justin Lidard, Lars L. Ankile, Anthony Simeonov, Pulkit Agrawal, Anirudha Majumdar, Benjamin Burchfiel, Hongkai Dai, Max Simchowitz. Diffusion Policy Policy Optimization. Princeton University; Massachusetts Institute of Technology; Toyota Research Institute; Carnegie Mellon University, 2024.
- [133] James L McClelland. Incorporating rapid neocortical learning of new schema-consistent information into complementary learning systems theory. Stanford University, 2013.
- [134] Ron Amit, Baruch Epstein, Shay Moran, Ron Meir. Integral Probability Metrics PAC-Bayes Bounds. Technion - Israel Institute of Technology; Google Research, 2022.
- [135] Anna Korba, François Portier. Adaptive Importance Sampling meets Mirror Descent: a Bias-variance tradeoff. CREST, ENSAE, Institut Polytechnique de Paris, ENSAI, 2021.
- [136] Michael J. Lew. A reckless guide to P-values. University of Melbourne, 2019.
- [137] Audun Jsang. Fission of Opinions in Subjective Logic. University of Oslo - UNIK Graduate Center, 2009.
- [138] John Miller, Juan C. Perdomo, Tijana Zrnic. Outside the Echo Chamber: Optimizing the Performative Risk. University of California, Berkeley, 2021.
- [139] Topi Halme, H. Vincent Poor, Visa Koivunen. Sequential Change Detection using Mixtures of Predictive Distributions. Aalto University, Princeton University, 2026.
- [140] Ke-Lin Du, Rengong Zhang, Bingchun Jiang, Jie Zeng, Jiabin Lu. Understanding Machine Learning Principles: Learning, Inference, Generalization, and Computational Learning Theory. Guangdong University of Science and Technology, Zhejiang Yugong Information Technology Co., Ltd., Shenzhen Feng Xing Tai Bao Technology Co., Ltd., Guangdong University of Technology., 2025.

- [141] L. W. Freriks, P. J. M. Cluitmans, M. J. van Gils. The Adaptive Resonance Theory network : (clustering-)behaviour in relation with Brainstem Auditory Evoked Potential patterns. Eindhoven University of Technology (TU/e), 1992.
- [142] J. M. Carlson, John Doyle. Highly Optimized Tolerance: A Mechanism for Power Laws in Designed Systems. University of California, Santa Barbara; California Institute of Technology (Caltech), 1999.
- [143] Nassim Nicholas Taleb. How Much Data Do You Need? A Pre-asymptotic Metric for Fat-tailedness. Tandon School of Engineering, New York University, 2018.
- [144] Omar Rivasplata, Emilio Parrado-Hernández, John Shawe-Taylor, Shiliang Sun, Csaba Szepesvári. PAC-Bayes bounds for stable algorithms with instance-dependent priors. UCL, University Carlos III of Madrid, East China Normal University, DeepMind, 2018.
- [145] Sham M. Kakade, Karthik Sridharan, Ambuj Tewari. On the Complexity of Linear Prediction: Risk Bounds, Margin Bounds, and Regularization. Toyota Technological Institute — Chicago, 2008.
- [146] I. Gibbs and E. Candès. Adaptive conformal inference under distribution shift. *Advances in Neural Information Processing Systems*, 34:1660–1672, 2021, 2021. <https://arxiv.org/abs/2106.00170>
- [147] R. C. Bradley. Basic properties of strong mixing conditions. A survey and some open questions. *Probability Surveys*, 2:107–144, 2005, 2005. <https://doi.org/10.1214/154957805100000104>
- [148] L. Paninski. Estimation of entropy and mutual information. *Neural Computation*, 15(6):1191–1253, 2003, 2003. <https://doi.org/10.1162/089976603321780272>
- [149] R. J. McCann. Existence and uniqueness of monotone measure-preserving maps. *Duke Mathematical Journal*, 80(2):307–324, 1995, 1995. <https://doi.org/10.1215/S0012-7094-95-08013-2>
- [150] R. Dahlhaus. Fitting time series models to nonstationary processes. *The Annals of Statistics*, 25(1):1–37, 1997, 1997. <https://doi.org/10.1214/aos/1034276620>
- [151] L. Ambrosio, N. Gigli, and G. Savaré. *Gradient Flows: in Metric Spaces and in the Space of Probability Measures*. Birkhäuser, 2008, 2008. <https://doi.org/10.1007/978-3-7643-8722-8>
- [152] I. Csiszár. I-divergence geometry of probability distributions and minimization problems. *The Annals of Probability*, 3(1):146–158, 1975, 1975. <https://doi.org/10.1214/aop/1176996454>
- [153] A. Bifet and R. Gavaldà. Learning from time-changing data with adaptive windowing. In *SIAM International Conference on Data Mining*, pages 443–448, 2007, 2007. <https://doi.org/10.1137/1.9781611972771.42>
- [154] J. Gama, P. Medas, G. Castillo, and P. Rodrigues. Learning with drift detection. In *Brazilian Symposium on Artificial Intelligence*, pages 286–295, 2004, 2004. https://doi.org/10.1007/978-3-540-28645-5_29

- [155] E. Hazan, A. Agarwal, and S. Kale. Logarithmic regret algorithms for online convex optimization. *Machine Learning*, 69:169–192, 2007, 2007. <https://doi.org/10.1007/s10994-007-5016-8>
- [156] D. S. Modha and E. Masry. Minimum complexity regression estimation with weakly dependent observations. *IEEE Transactions on Information Theory*, 42(6):2133–2145, 1996, 1996. <https://doi.org/10.1109/18.556602>
- [157] C. Villani. *Optimal Transport: Old and New*. Springer Science & Business Media, volume 338, 2009, 2009. <https://doi.org/10.1007/978-3-540-71050-9>
- [158] B. Yu. Rates of convergence for empirical processes of stationary mixing sequences. *The Annals of Probability*, 22(1):94–116, 1994, 1994. <https://doi.org/10.1214/aop/1176988849>
- [159] Q. Liu and D. Wang. Stein variational gradient descent: A general purpose Bayesian inference algorithm. In *Advances in Neural Information Processing Systems*, volume 29, pages 2378–2386, 2016, 2016. <https://arxiv.org/abs/1608.04471>
- [160] R. Jordan, D. Kinderlehrer, and F. Otto. The variational formulation of the Fokker–Planck equation. *SIAM Journal on Mathematical Analysis*, 29(1):1–17, 1998, 1998. <https://doi.org/10.1137/S0036141096303359>
- [161] J. Dambre, D. Verstraeten, B. Schrauwen, S. Massar. Information processing capacity of dynamical systems. *Scientific Reports*, 2:514, 2012, 2012. <https://doi.org/10.1038/srep00514>
- [162] R. Landauer. Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development*, 5(3):183–191, 1961, 1961. <https://doi.org/10.1147/rd.53.0183>
- [163] C. Godreche, J.-M. Luck. Characterising the nonequilibrium stationary states of Ornstein-Uhlenbeck processes. *Journal of Physics A: Mathematical and Theoretical*, 52(3):035002, 2018, 2018. <https://arxiv.org/abs/1807.00694>
- [164] I. Prigogine. *Time, Structure and Fluctuations*. Nobel Lecture, Premio Nobel de Quimica, 1977. <https://www.nobelprize.org/prizes/chemistry/1977/prigogine/lecture/>
- [165] T. M. Cover, J. A. Thomas. *Elements of Information Theory*. Wiley-Interscience, 2a edicao, 2006. <https://doi.org/10.1002/047174882X>
- [166] V. I. Oseledets. A multiplicative ergodic theorem: Lyapunov characteristic numbers for dynamical systems. *Transactions of the Moscow Mathematical Society*, 19:197–231, 1968, 1968. <https://doi.org/10.1090/trans2/019>
- [167] G. Benettin, L. Galgani, A. Giorgilli, J.-M. Strelcyn. Lyapunov characteristic exponents for smooth dynamical systems and for Hamiltonian systems - a method for computing all of them. *Meccanica*, 15(1):9–20, 1980, 1980. <https://doi.org/10.1007/BF02128236>
- [168] L. N. Trefethen, M. Embree. *Spectra and Pseudospectra: The Behavior of Nonnormal Matrices and Operators*. Princeton University Press, 2005, 2005. <https://doi.org/10.1515/9781400830473>

- [169] R. A. Horn, C. R. Johnson. Matrix Analysis. Cambridge University Press, 2a edicao, 2013, 2013. <https://doi.org/10.1017/CB09781139020411>
- [170] J. D. Lee, I. Panageas, G. Piliouras, M. Simchowitz, M. I. Jordan, B. Recht. First-order methods almost always avoid strict saddle points. Mathematical Programming, 176(1-2):311–337, 2019, 2019. <https://doi.org/10.1007/s10107-019-01374-3>
- [171] N. J. Higham. Accuracy and Stability of Numerical Algorithms. SIAM, 2a edicao, 2002, 2002. <https://doi.org/10.1137/1.9780898718027>
- [172] G. Gallavotti, E. G. D. Cohen. Dynamical ensembles in stationary states. Journal of Statistical Physics, 80(5-6):931–970, 1995, 1995. <https://doi.org/10.1007/BF02179860>
- [173] S. Ghosh. Non-normal spectral signatures of instability in neural network training dynamics. arXiv:2605.23476, 2026. <https://arxiv.org/abs/2605.23476>
- [174] A. H. Ariq. Pseudospectral Bounds for Transient Amplification in Coupled Gradient Descent. arXiv:2606.04031, 2026. <https://arxiv.org/abs/2606.04031>
- [175] R. J. Leveque, L. N. Trefethen. On the resolvent condition in the Kreiss Matrix Theorem. BIT Numerical Mathematics, 24(5):584–591, 1984, 1984. <https://doi.org/10.1007/BF01934916>
- [176] F. L. Bauer, C. T. Fike. Norms and exclusion theorems. Numerische Mathematik, 2(1):137–141, 1960, 1960. <https://doi.org/10.1007/BF01386217>
- [177] P. Henrici. Bounds for iterates, inverses, spectral variation and fields of values of non-normal matrices. Numerische Mathematik, 4(1):24–40, 1962, 1962. <https://doi.org/10.1007/BF01386294>
- [178] L. N. Trefethen, A. E. Trefethen, S. C. Reddy, T. A. Driscoll. Hydrodynamic stability without eigenvalues. Science, 261(5121):578–584, 1993, 1993. <https://doi.org/10.1126/science.261.5121.578>
- [179] B. T. Polyak. Some methods of speeding up the convergence of iteration methods. USSR Computational Mathematics and Mathematical Physics, 4(5):1–17, 1964, 1964. [https://doi.org/10.1016/0041-5553\(64\)90137-5](https://doi.org/10.1016/0041-5553(64)90137-5)
- [180] G. H. Golub, C. F. Van Loan. Matrix Computations. Johns Hopkins University Press, 4a edicao, 2013, 2013. <https://doi.org/10.56021/9781421407944>
- [181] N. J. Higham. Functions of Matrices: Theory and Computation. SIAM, Philadelphia, 2008, 2008. <https://doi.org/10.1137/1.9780898717778>
- [182] I. C. F. Ipsen. Computing an eigenvector with inverse iteration. SIAM Review, 39(2):254–291, 1997, 1997. <https://doi.org/10.1137/S0036144596300773>
- [183] I. Goldhirsch, P.-L. Sulem, S. A. Orszag. Stability and Lyapunov stability of dynamical systems: a differential approach and a numerical method. Physica D: Nonlinear Phenomena, 27(3):311–337, 1987, 1987. [https://doi.org/10.1016/0167-2789\(87\)90034-0](https://doi.org/10.1016/0167-2789(87)90034-0)
- [184] A. Pikovsky, A. Politi. Lyapunov Exponents: A Tool to Explore Complex Dynamics. Cambridge University Press, 2016, 2016. <https://doi.org/10.1017/CB09781139343473>

- [185] W. D. Heiss. The physics of exceptional points. *Journal of Physics A: Mathematical and Theoretical*, 45(44):444016, 2012, 2012. <https://doi.org/10.1088/1751-8113/45/44/444016>
- [186] T. Kato. *Perturbation Theory for Linear Operators*. Springer, Classics in Mathematics, 1995, 1995. <https://doi.org/10.1007/978-3-642-66282-9>
- [187] M. V. Berry. Physics of nonhermitian degeneracies. *Czechoslovak Journal of Physics*, 54(10):1039–1047, 2004, 2004. <https://doi.org/10.1023/B:CJOP.0000044002.05657.04>
- [188] L. N. Trefethen. Pseudospectra of linear operators. *SIAM Review*, 39(3):383–406, 1997, 1997. <https://doi.org/10.1137/S0036144595295284>
- [189] Y. Bengio, N. Leonard, A. Courville. Estimating or Propagating Gradients Through Stochastic Neurons for Conditional Computation. *arXiv:1308.3432*, 2013. <https://arxiv.org/abs/1308.3432>
- [190] G. Kerg, K. Goyette, M. Puelma Touzel, G. Gidel, E. Vorontsov, Y. Bengio, G. Lajoie. Non-normal Recurrent Neural Network (nnRNN): learning long time dependencies while improving expressivity with transient dynamics. *NeurIPS 2019*; *arXiv:1905.12080*, 2019. <https://arxiv.org/abs/1905.12080>
- [191] J. D. Botha. Products of diagonalizable matrices. *Linear Algebra and its Applications*, 273:65–82, 1998. [https://doi.org/10.1016/S0024-3795\(97\)00344-3](https://doi.org/10.1016/S0024-3795(97)00344-3)
- [192] H. Flanders. Elementary divisors of AB and BA . *Proceedings of the American Mathematical Society*, 2:871–874, 1951. <https://doi.org/10.1090/S0002-9939-1951-0044493-X>
- [193] R. A. Lippert, G. Strang. The Jordan forms of AB and BA . *Electronic Journal of Linear Algebra*, 18, 2009. <https://doi.org/10.13001/1081-3810.1313>
- [194] A. Eskin, C. Matheus. Semisimplicity of the Lyapunov spectrum for irreducible cocycles. *Israel Journal of Mathematics*, 230:973–1005, 2019. <https://doi.org/10.1007/s11856-019-1841-2>
- [195] J. Bochi, P. Lessa. On the distribution of the angle between Oseledets spaces. *Annales Henri Lebesgue*, 9:65–84, 2026. <https://doi.org/10.5802/ahl.260>
- [196] W. M. Goldman. An exposition of results of Fricke. *arXiv:math/0402103*, 2004. <https://arxiv.org/abs/math/0402103>
- [197] R. Fricke. *Vorlesungen über die Theorie der automorphen Functionen*, Band 1. B. G. Teubner, Leipzig, 1897. <https://archive.org/details/vorlesungenber01fricuoft>
- [198] A. F. Beardon. *The Geometry of Discrete Groups*. Graduate Texts in Mathematics 91, Springer, 1983. <https://doi.org/10.1007/978-1-4612-1146-4>
- [199] Niklas Pfister, Peter Bühlmann, Bernhard Schölkopf, Jonas Peters. Kernel-Based Tests for Joint Independence. *Journal of the Royal Statistical Society Series B*, 80(1):5–31, 2018. <https://doi.org/10.1111/rssb.12235>

- [200] Gábor J. Székely, Maria L. Rizzo, Nail K. Bakirov. Measuring and testing dependence by correlation of distances. *The Annals of Statistics*, 35(6):2769-2794, 2007. <https://doi.org/10.1214/009053607000000505>
- [201] Abraham Wald. Sequential Tests of Statistical Hypotheses. *The Annals of Mathematical Statistics*, 16(2):117-186, 1945. <https://doi.org/10.1214/aoms/1177731118>
- [202] Robert E. Kass, Adrian E. Raftery. Bayes Factors. *Journal of the American Statistical Association*, 90(430):773-795, 1995. <https://doi.org/10.1080/01621459.1995.10476572>
- [203] Vladimir N. Vapnik, Alexey Ya. Chervonenkis. On the Uniform Convergence of Relative Frequencies of Events to Their Probabilities. *Theory of Probability & Its Applications*, 16(2):264-280, 1971. <https://doi.org/10.1137/1116025>
- [204] Bradley Efron. Bootstrap Methods: Another Look at the Jackknife. *The Annals of Statistics*, 7(1):1-26, 1979. <https://doi.org/10.1214/aos/1176344552>
- [205] John P. A. Ioannidis. Why Most Published Research Findings Are False. *PLoS Medicine*, 2(8):e124, 2005. <https://doi.org/10.1371/journal.pmed.0020124>
- [206] Ronald L. Wasserstein, Nicole A. Lazar. The ASA Statement on p-Values: Context, Process, and Purpose. *The American Statistician*, 70(2):129-133, 2016. <https://doi.org/10.1080/00031305.2016.1154108>
- [207] Brandon Amos, Lei Xu, J. Zico Kolter. Input Convex Neural Networks. *Proceedings of the 34th International Conference on Machine Learning (PMLR 70:316-325)*, 2017. <https://proceedings.mlr.press/v70/amos17a.html>
- [208] Wolfgang Maass, Thomas Natschläger, Henry Markram. Real-Time Computing Without Stable States: A New Framework for Neural Computation Based on Perturbations. *Neural Computation*, 14(11):2531-2560, 2002. <https://doi.org/10.1162/089976602760407955>
- [209] Arthur P. Dempster. Upper and Lower Probabilities Induced by a Multivalued Mapping. *The Annals of Mathematical Statistics*, 38(2):325-339, 1967. <https://doi.org/10.1214/aoms/1177698950>
- [210] Audun Jsang. *Subjective Logic: A Formalism for Reasoning Under Uncertainty*. Springer (Artificial Intelligence: Foundations, Theory, and Algorithms), 2016. <https://doi.org/10.1007/978-3-319-42337-1>
- [211] Noga Alon, Yossi Matias, Mario Szegedy. The Space Complexity of Approximating the Frequency Moments. *Journal of Computer and System Sciences*, 58(1):137-147, 1999. <https://doi.org/10.1006/jcss.1997.1545>
- [212] Piotr Indyk, David P. Woodruff. Optimal Approximations of the Frequency Moments of Data Streams. *Proceedings of the 37th Annual ACM Symposium on Theory of Computing (STOC)*, 202-208, 2005. <https://doi.org/10.1145/1060590.1060621>
- [213] David P. Woodruff. Sketching as a Tool for Numerical Linear Algebra. *Foundations and Trends in Theoretical Computer Science*, 10(1-2):1-157, 2014. <https://doi.org/10.1561/04000000060>

- [214] Celestine Mender-Dünner, Juan Perdomo, Tijana Zrnic, Moritz Hardt. Stochastic Optimization for Performative Prediction. Proceedings of the 37th International Conference on Machine Learning (PMLR 119:7081-7090), 2020. <https://proceedings.mlr.press/v119/mender-dunner20a.html>
- [215] T. Higuchi. Approach to an Irregular Time Series on the Basis of the Fractal Theory. *Physica D: Nonlinear Phenomena*, 31(2):277-283, 1988. [https://doi.org/10.1016/0167-2789\(88\)90081-4](https://doi.org/10.1016/0167-2789(88)90081-4)
- [216] Israel Gohberg, Peter Lancaster, Leiba Rodman. Invariant Subspaces of Matrices with Applications. Society for Industrial and Applied Mathematics (SIAM Classics in Applied Mathematics 51), 2006. <https://doi.org/10.1137/1.9780898719093>
- [217] Philippe Bougerol, Jean Lacroix. Products of Random Matrices with Applications to Schrödinger Operators. Birkhäuser Boston (Progress in Probability and Statistics 8), 1985. <https://doi.org/10.1007/978-1-4684-9172-2>
- [218] Per Bak, Chao Tang, Kurt Wiesenfeld. Self-organized Criticality: An Explanation of $1/f$ Noise. *Physical Review Letters*, 59(4):381-384, 1987. <https://doi.org/10.1103/PhysRevLett.59.381>
- [219] Sergey V. Buldyrev, Roni Parshani, Gerald Paul, H. Eugene Stanley, Shlomo Havlin. Catastrophic Cascade of Failures in Interdependent Networks. *Nature*, 464:1025-1028, 2010. <https://doi.org/10.1038/nature08932>
- [220] Marten Scheffer, Jordi Bascompte, William A. Brock, Victor Brovkin, Stephen R. Carpenter, Vasilis Dakos, Hermann Held, Egbert H. van Nes, Max Rietkerk, George Sugihara. Early-warning signals for critical transitions. *Nature*, 461:53-59, 2009. <https://doi.org/10.1038/nature08227>
- [221] Vasilis Dakos, Stephen R. Carpenter, William A. Brock, Aaron M. Ellison, Vishwesha Guttal, Anthony R. Ives, Sonia Kéfi, David A. Seekell, and others. Methods for Detecting Early Warnings of Critical Transitions in Time Series Illustrated Using Simulated Ecological Data. *PLoS ONE*, 7(7):e41010, 2012. <https://doi.org/10.1371/journal.pone.0041010>
- [222] Vivek S. Borkar. Stochastic approximation with two time scales. *Systems & Control Letters*, 29(5):291-294, 1997. [https://doi.org/10.1016/S0167-6911\(97\)00015-1](https://doi.org/10.1016/S0167-6911(97)00015-1)
- [223] Richard S. Sutton, Doina Precup, Satinder Singh. Between MDPs and semi-MDPs: A framework for temporal abstraction in reinforcement learning. *Artificial Intelligence*, 112(1-2):181-211, 1999. [https://doi.org/10.1016/S0004-3702\(99\)00052-1](https://doi.org/10.1016/S0004-3702(99)00052-1)
- [224] Andrew Y. Ng, Daishi Harada, Stuart Russell. Policy invariance under reward transformations: Theory and application to reward shaping. Proceedings of the 16th International Conference on Machine Learning, 278-287, 1999. <https://people.eecs.berkeley.edu/~russell/papers/ml99-shaping.pdf>
- [225] Herbert Jaeger. The echo state approach to analysing and training recurrent neural networks. GMD Report 148, German National Research Center for Information Technology, 2001. <https://www.ai.rug.nl/minds/uploads/EchoStatesTechRep.pdf>

- [226] Allen G. Hart, James L. Hook, Jonathan H. P. Dawes. Embedding and approximation theorems for echo state networks. *Neural Networks*, 128:234-247, 2020. <https://doi.org/10.1016/j.neunet.2020.05.006>
- [227] M. U. Gaimann et al.. Optimal information injection and transfer mechanisms for active matter reservoir computing. *arXiv:2509.01799*, 2025. <https://arxiv.org/abs/2509.01799>
- [228] Édgar Roldán, Izaak Neri, Meinolf Roldán, Jürgen Mehl, Frank Jülicher. Statistics of Infima and Stopping Times of Entropy Production and Applications to Active Molecular Processes. *Physical Review X*, 7(2):021026, 2017. <https://doi.org/10.1103/PhysRevX.7.021026>
- [229] Yoshihiko Hasegawa. Uncertainty relations in stochastic processes: An information inequality approach. *Physical Review E*, 99(6):062126, 2019. <https://doi.org/10.1103/PhysRevE.99.062126>
- [230] Jean-David Benamou, Yann Brenier. A computational fluid mechanics solution to the Monge-Kantorovich mass transfer problem. *Numerische Mathematik*, 84(3):375-393, 2000. <https://doi.org/10.1007/s002110050002>
- [231] Gunnar Blom. Some properties of incomplete U-statistics. *Biometrika*, 63(3):573-580, 1976. <https://doi.org/10.1093/biomet/63.3.573>
- [232] Xiaoming Huo, Gábor J. Székely. Fast Computing for Distance Covariance. *Technometrics*, 58(4):435-447, 2016. <https://doi.org/10.1080/00401706.2015.1054435>
- [233] I. Richard Savage. Mills' ratio for multivariate normal distributions. *Journal of Research of the National Bureau of Standards B*, 66(3):93-96, 1962. https://nvlpubs.nist.gov/nistpubs/jres/66B/jresv66n3p93_A1b.pdf
- [234] Enkelejd Hashorva, Jürg Hüsler. On multivariate Gaussian tails. *Annals of the Institute of Statistical Mathematics*, 55(3):507-522, 2003. <https://doi.org/10.1007/BF02530509>